

ABHIPRIYO BISWAS

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PROFESSIONAL SUMMARY

Software Engineer with 7+ years of experience spanning **full-stack web development**, **backend/API engineering**, and **mobile application development**, and holds an **MSc in Data Science (CGPA: 7.78)**, completed alongside this engineering career. Career progression from Web Designer to Frontend Developer to Full Stack Developer, with hands-on production experience across **React.js**, **AngularJS**, **React Native**, **Node.js**, **Python**, and **FastAPI**, building and maintaining downstream APIs, state management (Redux), caching (Redis), and secure credential management (GCP Secret Manager) for high-traffic enterprise applications. Additionally skilled in end-to-end **ETL/ELT data pipelines** (Python, PostgreSQL, Apache Airflow, dbt), **interactive dashboards** (Plotly, Dash, Power BI), and **machine learning models** (Scikit-learn, XGBoost, TensorFlow, PyTorch), including a deep learning-based breast cancer detection system. Comfortable owning work end-to-end — from production debugging and unit testing to data pipeline design and model evaluation — within Agile, cross-functional, and distributed teams.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript (ES6+), TypeScript, SQL, Bash

Frontend: React.js, AngularJS, Angular, React Native, Next.js, Redux, Redux-Thunk, HTML5, CSS3, Bootstrap, jQuery, Responsive Design, Component-Based Architecture

Backend & APIs: Node.js, Python, FastAPI, REST API Design & Integration, Layered API Architecture, JSON, HTTP, Redis (Caching)

Data Engineering: ETL/ELT Pipelines, dbt, Apache Airflow, Data Modeling, Data Warehousing, Batch Processing, Data Transformation & Validation, Idempotent Upserts, Pipeline Orchestration

Data Analysis & Visualisation: Exploratory Data Analysis (EDA), Data Cleaning, Feature Engineering, Pandas, NumPy, Matplotlib, Plotly, Dash, Power BI (familiar), KPI Design, Reporting

Machine Learning: Scikit-learn, XGBoost, Classification, Regression, Clustering, SMOTE, Cross-Validation, Hyperparameter Tuning, Model Evaluation (ROC-AUC, F1, Precision/Recall), TensorFlow, PyTorch, CNN (Medical Imaging), Transfer Learning

Databases & SQL: PostgreSQL, MySQL, Advanced SQL (Window Functions, CTEs, Joins, Aggregations, Subqueries, Indexing, Query Optimisation)

Cloud & DevOps: AWS (S3, Glue), GCP (BigQuery, Secret Manager, Cloud Logging), Azure (Fundamentals), Docker, Git, GitHub, Jenkins (CI)

Testing & Quality: Unit Testing, Jest, Enzyme, React Testing Library, Debugging, Production Issue Resolution, Error Handling, Logging

Big Data & Orchestration: Apache Airflow, Apache Spark (familiar)

Statistical Analysis: Hypothesis Testing, Descriptive Statistics, Distribution Analysis, Resampling Methods

Tools: VS Code, Jupyter Notebook, Overleaf, Linux

Methodologies: Agile, SDLC, Modular Architecture, End-to-End Project Delivery, Cross-functional & Distributed Team Collaboration, Documentation

PROFESSIONAL EXPERIENCE

Career Progression: Web Designer → Frontend Developer → Full Stack Developer

Full Stack Developer / Contractor

Aug 2021 – Dec 2025

TELUS Digital (via Etelligens Technologies)

Remote, Noida, India

- Worked as a Contractor on the **Sapphire team**, contributing to **CASA**, a customer support application used by call center agents to access customer profiles, billing information, service plans, offers, and account dashboards in real time.
- Developed and enhanced features in a **React.js** and **Node.js** application, integrating secure backend services to retrieve and display offers data from downstream APIs; used **Redux** and **Redux-Thunk** to consume and manage backend API responses.
- Collaborated with API teams to implement a **layered API architecture**, where frontend services consumed intermediary APIs that securely accessed and aggregated data from backend systems.

- On the backend, consumed downstream APIs and performed data sanitization (Node.js/JavaScript) to return only relevant data to the frontend.
- Wrote **unit tests** using **React Testing Library** and **Jenkins** for frontend, and **Jest** for backend; used **Redis** for caching and **GCP Secret Manager** for secure credential storage.
- Monitored **GCP Logs** for errors and warnings; **debugged and resolved production issues** faced by customer care agents in real time during live customer calls.
- Participated in the organization's transition from monolithic architecture to microservices, decomposing application modules into independently deployable services to improve scalability and maintainability.
- Later joined the **API development team**, designing and maintaining downstream APIs using **Python**, **FastAPI**, and **PostgreSQL**, supporting data integration, business workflows, and enterprise applications; optimized API performance, backend processing logic, and database queries, improving response times and system reliability.
- Worked within Agile teams using Git-based version control, contributing to production releases and maintaining high service stability across customer-facing systems.

Frontend Developer

Dec 2020 – Dec 2025

Etelligens Technologies

Remote, Noida, India

- Maintained and shipped new features for **TravelNet**, a hybrid mobile application, and **Alloodo**, a game-rental client website (frontend maintenance and feature enhancement).
- Developed production-grade **React.js** and **AngularJS** applications, consuming backend REST APIs and optimizing data-driven UI performance across 5+ client projects, including **Pyro**, **Local Kings**, and **Huskarma**.
- Collaborated with distributed engineering teams in remote environments, consistently delivering features on schedule across Agile sprint cycles.

Mobile Application Developer

Jun 2020 – Dec 2020

Globiz Technology Inc.

Remote, Ludhiana, India

- Member of both the design and development teams; delivered fully responsive, cross-browser-tested, pixel-perfect static website designs from mockups, and pixel-perfect mobile app screens from UI mockups in **React Native**.
- Developed hybrid mobile applications using **React Native**, consuming RESTful APIs and integrating structured backend data into performant mobile UIs, including **LocalCart** and **LineManagement** (deployed to Google Play Store).

Web Designer / Associate Frontend Developer

Aug 2018 – May 2020

Promatics Technologies Pvt. Ltd.

Ludhiana, India

- Promoted from **Web Designer** to **Associate Frontend Developer** within one year in recognition of strong technical performance, rapid skill development, and consistent project contributions.
- Designed responsive website mockups, mobile application UI mockups, and branding assets (logos, banners, marketing graphics) using **Adobe Photoshop** and **Adobe Illustrator**.
- Converted approved UI/UX designs into responsive, cross-browser compatible web interfaces using **HTML5**, **CSS3**, **JavaScript/jQuery**, and **Bootstrap**.
- Collaborated with designers, developers, and project managers to deliver multiple client-facing websites from design through frontend implementation.
- Contributed to frontend feature development, UI enhancements, bug fixes, and production-ready web pages across multiple projects.
- Built and maintained reusable UI components while ensuring responsive design and a consistent user experience across desktop and mobile devices.
- **Selected Portfolio:** UCatch, PortaPrep, Babzer, GPS Colombia (Design).

DATA ENGINEERING & MACHINE LEARNING PROJECTS

Breast Cancer Detection — MSc Dissertation | *Python, CNN, TensorFlow, PyTorch, Medical Imaging, Statistical Analysis* 2025 – 2026

- Developed a deep learning classification model for breast cancer detection using histopathological image data as part of MSc dissertation research.
- Applied Convolutional Neural Networks (CNNs) with transfer learning to classify malignant vs. benign tissue samples, achieving strong diagnostic performance metrics.
- Conducted statistical analysis of model outputs including ROC-AUC, precision/recall, and confusion matrix evaluation to ensure clinical-grade reliability.
- Addressed class imbalance through data augmentation and resampling strategies, improving minority-class recall.
- Produced detailed research documentation covering methodology, experimental design, results analysis, and conclusions — demonstrating end-to-end project ownership.

CoinPulse — Crypto Market ETL Platform | *Python, PostgreSQL, Apache Airflow, dbt, Docker, REST API*, **2026**
Data Warehousing [Source Code]

- Designed and deployed a production-grade end-to-end ETL/ELT data pipeline ingesting cryptocurrency market data from the CoinGecko REST API into a PostgreSQL data warehouse.
- Automated daily extraction and transformation of market data from raw JSON payloads into structured warehouse tables, eliminating manual data collection and processing.
- Wrote dbt transformation models to convert raw ingested data into structured Silver and Gold layer tables, with built-in dbt tests for data quality enforcement.
- Implemented idempotent upsert logic and primary-key conflict handling to prevent duplicate records and maintain data integrity across repeated pipeline executions.
- Orchestrated pipeline scheduling and workflow automation using Apache Airflow DAGs within a Dockerized environment.
- Built SQL analytics reports to surface historical price trends, asset performance KPIs, and market movement insights across 100+ cryptocurrency assets.
- Designed a modular Python architecture with reusable components and centralized configuration management, improving maintainability and reducing code duplication.
- Implemented validation checks and error handling mechanisms to ensure reliable ingestion and transformation of API data.

LogPulse — Application Log Analytics Pipeline | *Python, PostgreSQL, Batch Processing, Data Modeling, SQL* **2026**
Analytics [Source Code]

- Built a scalable batch data pipeline to ingest and process synthetic application logs, transforming unstructured raw text into structured relational records in PostgreSQL.
- Implemented data parsing, field extraction, and data validation checks to ensure clean, analytics-ready datasets.
- Developed SQL-based analytical reports and KPI dashboards to monitor request volume, HTTP status distribution, and application error trends.
- Engineered scalable ingestion design patterns supporting high-volume log data with built-in data quality enforcement at ingestion.
- Designed normalized relational schemas and indexing strategies to support efficient querying and reporting workloads.

Interactive Movie Analytics Dashboard | *Python, Pandas, NumPy, Plotly, Dash* [Source Code] **2025**

- Built an interactive dashboard to analyze movie revenue, ratings, trends, and performance metrics.
- Created dynamic visualizations including KPIs, bar charts, trend analysis, and distribution plots.
- Performed full data preprocessing including cleaning, deduplication, and metric creation.
- Implemented user-driven filters and deployed the dashboard using a production-ready structure.

Credit Card Fraud Detection — ML Data Pipeline | *Python, Pandas, NumPy, Scikit-learn, XGBoost, Feature Engineering, Data Preprocessing* [Source Code] **2025**

- Built end-to-end data preprocessing and feature engineering pipelines on a transactional dataset containing over 1M+ records.
- Applied data cleaning, normalization, and SMOTE-based resampling techniques to address class imbalance and improve training data quality.
- Developed automated transformation workflows producing standardized, model-ready datasets with validated feature distributions and schema consistency, compatible with Scikit-learn and XGBoost estimators.
- Applied cross-validation and performance benchmarking, deriving actionable insights on transaction risk patterns.

EDUCATION

Master of Science in Data Science **Jul 2024 – Jun 2026**

Central University of Haryana, Mahendargarh *CGPA: 7.78 — Completed; Degree Certificate Awaited*

- Data Engineering, Big Data Systems, Database Management, Machine Learning, Statistical Analysis, Data Visualisation.
- Dissertation: Breast Cancer Detection using Deep Learning and Medical Image Classification

Bachelor of Arts in Computer Science **Dec 2020 – Apr 2024**

Lovely Professional University

CERTIFICATIONS

HackerRank Skill Certifications: **Python (Basic), SQL (Basic), SQL (Intermediate)**

ADDITIONAL INFORMATION

Languages: English, Hindi, Punjabi, Bengali

Open to: On-site, Remote, and Hybrid positions

Target Roles: Python Backend Developer, Full Stack Developer, Data Analyst, Data Engineer, Data Scientist